

bootstrap container layout
what is bootstrap
using bootstrap
button component
explain
badge all concept explain with code
alert component
button group component
navbar component
card component
grid component
grid vertical alignment
form control bootstrap
select in form
checkbox and radio
checkbox and radio in form
form layout

Bootstrap Comprehensive Guide

What is Bootstrap?

Bootstrap is a popular open-source front-end framework for developing responsive, mobile-first websites. It contains CSS and JavaScript-based design templates for typography, forms, buttons, navigation, and other interface components.

Key Features:

- **Responsive Grid System** - Adapts to all screen sizes
- **Pre-styled Components** - Ready-to-use UI elements
- **Utility Classes** - For quick styling without custom CSS
- **Cross-browser Compatibility**
- **Mobile-first Approach**

1. Bootstrap Container Layout

Basic Structure:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <!-- Bootstrap CSS -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.c
ss" rel="stylesheet">
  <title>Bootstrap Guide</title>
</head>
<body>
  <!-- Container Examples -->
  <div class="container">Fixed width container</div>
  <div class="container-fluid">Full width container</div>

  <!-- Bootstrap JS Bundle -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundl
e.min.js"></script>
</body>
</html>
```

Container Types:

```
<div class="container my-3 p-3 bg-light">
  <h4>.container</h4>
  <p>Fixed width at each breakpoint</p>
</div>
html
<div class="container-sm my-3 p-3 bg-light">
  <h4>.container-sm</h4>
  <p>100% wide until small breakpoint</p>
</div>

<div class="container-lg my-3 p-3 bg-light">
```

```

    <h4>.container-lg</h4>
    <p>100% wide until large breakpoint</p>
  </div>

  <div class="container-fluid my-3 p-3 bg-light">
    <h4>.container-fluid</h4>
    <p>Always 100% width</p>
  </div>

```

2. Button Component

Basic Buttons:

```

html
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-link">Link</button>

<!-- Outline Buttons -->
<button type="button" class="btn btn-outline-primary">Outline Primary</button>
<button type="button" class="btn btn-outline-success">Outline Success</button>

<!-- Button Sizes -->
<button type="button" class="btn btn-primary btn-lg">Large</button>
<button type="button" class="btn btn-primary">Regular</button>
<button type="button" class="btn btn-primary btn-sm">Small</button>

<!-- Disabled State -->
<button type="button" class="btn btn-primary" disabled>Disabled</button>

```

3. Badge Component

Basic Badges:

```

<h1>Heading <span class="badge bg-primary">New</span></h1>
<h2>Heading <span class="badge bg-secondary">New</span></h2>
html

<!-- Contextual Badges -->
<span class="badge bg-primary">Primary</span>
<span class="badge bg-secondary">Secondary</span>
<span class="badge bg-success">Success</span>
<span class="badge bg-danger">Danger</span>
<span class="badge bg-warning text-dark">Warning</span>
<span class="badge bg-info text-dark">Info</span>
<span class="badge bg-light text-dark">Light</span>
<span class="badge bg-dark">Dark</span>

<!-- Pill Badges -->
<span class="badge rounded-pill bg-primary">Primary Pill</span>
<span class="badge rounded-pill bg-success">Success Pill</span>

<!-- Badge in Button -->
<button type="button" class="btn btn-primary">
  Notifications <span class="badge bg-danger">4</span>
</button>

<!-- Positioned Badges -->
<button type="button" class="btn btn-primary position-relative">
  Inbox
  <span class="position-absolute top-0 start-100 translate-middle badge rounded-pill bg-danger">
    99+
    <span class="visually-hidden">unread messages</span>
  </span>
</button>

```

4. Alert Component

Basic Alerts:

```
html
```

```

<!-- Basic Alerts -->
<div class="alert alert-primary" role="alert">
  A simple primary alert
</div>

html
  A simple success alert
</div>
<div class="alert alert-danger" role="alert">
  A simple danger alert
</div>

<!-- Alert with Link -->
<div class="alert alert-primary" role="alert">
  A simple primary alert with <a href="#" class="alert-link">an example link</a>
</div>

<!-- Dismissible Alert -->
<div class="alert alert-warning alert-dismissible fade show" role="alert">
  <strong>Holy guacamole!</strong> You can close this alert.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>

<!-- Alert with Content -->
<div class="alert alert-success" role="alert">
  <h4 class="alert-heading">Well done!</h4>
  <p>Aww yeah, you successfully read this important alert message.</p>
  <hr>
  <p class="mb-0">Whenever you need to, be sure to use margin utilities.</p>
</div>

```

5. Button Group Component

Button Groups:

```

html

<!-- Basic Button Group -->
<div class="btn-group" role="group">
  <button type="button" class="btn btn-primary">Left</button>
  <button type="button" class="btn btn-primary">Middle</button>
  <button type="button" class="btn btn-primary">Right</button>

```

```
</div>
```

```
<!-- Button Toolbar -->
```

```
<div class="btn-toolbar" role="toolbar">
  <div class="btn-group me-2">
    <button class="btn btn-secondary">1</button>
    <button class="btn btn-secondary">2</button>
    <button class="btn btn-secondary">3</button>
  </div>
  <div class="btn-group">
    <button class="btn btn-secondary">4</button>
    <button class="btn btn-secondary">5</button>
  </div>
</div>
```

```
<!-- Vertical Button Group -->
```

```
<div class="btn-group-vertical">
  <button type="button" class="btn btn-primary">Top</button>
  <button type="button" class="btn btn-primary">Middle</button>
  <button type="button" class="btn btn-primary">Bottom</button>
</div>
```

```
<!-- Button Group with Dropdown -->
```

```
<div class="btn-group">
  <button type="button" class="btn btn-primary">Action</button>
  <button type="button" class="btn btn-primary dropdown-toggle dropdown-toggle-split"
    data-bs-toggle="dropdown"></button>
  <ul class="dropdown-menu">
    <li><a class="dropdown-item" href="#">Action</a></li>
    <li><a class="dropdown-item" href="#">Another action</a></li>
  </ul>
</div>
```

6. Navbar Component

Navigation Bar:

```
html
```

```
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
  <div class="container-fluid">
```

```

<!-- Brand -->
<a class="navbar-brand" href="#">MySite</a>

<!-- Toggle Button -->
<button class="navbar-toggler" type="button" data-bs-toggle="collapse"
        data-bs-target="#navbarNav">
    <span class="navbar-toggler-icon"></span>
</button>

<!-- Navbar Items -->
<div class="collapse navbar-collapse" id="navbarNav">
    <ul class="navbar-nav">
        <li class="nav-item">
            <a class="nav-link active" href="#">Home</a>
        </li>
        <li class="nav-item">
            <a class="nav-link" href="#">Features</a>
        </li>
        <li class="nav-item dropdown">
            <a class="nav-link dropdown-toggle" href="#" data-bs-toggle="drop
down">
                Dropdown
            </a>
            <ul class="dropdown-menu">
                <li><a class="dropdown-item" href="#">Action</a></li>
                <li><a class="dropdown-item" href="#">Another action</a></li>
            </ul>
        </li>
    </ul>

    <!-- Search Form -->
    <form class="d-flex ms-auto">
        <input class="form-control me-2" type="search" placeholder="Search">
        <button class="btn btn-outline-success" type="submit">Search</button>
    </form>
</div>
</div>
</nav>

<!-- Color Variants -->
<nav class="navbar navbar-expand-lg navbar-light bg-primary">
    <!-- Content -->
</nav>

```

```
<nav class="navbar navbar-expand-lg navbar-dark bg-success">
  <!-- Content -->
</nav>
```

7. Card Component

Basic Card:

html

```
<div class="card" style="width: 18rem;">
  
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">Some quick example text to build on the card title.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
</div>

<!-- Card with Header and Footer -->
<div class="card">
  <div class="card-header">
    Featured
  </div>
  <div class="card-body">
    <h5 class="card-title">Special title treatment</h5>
    <p class="card-text">With supporting text below as a natural lead-in.</p>
    <a href="#" class="btn btn-primary">Go somewhere</a>
  </div>
  <div class="card-footer text-muted">
    2 days ago
  </div>
</div>

<!-- Card Groups -->
<div class="card-group">
  <div class="card">
    <div class="card-body">
      <h5 class="card-title">Card 1</h5>
      <p class="card-text">Content for card 1.</p>
    </div>
  </div>
```



```

<div class="card">
  <div class="card-body">
    <h5 class="card-title">Card 2</h5>
    <p class="card-text">Content for card 2.</p>
  </div>
</div>
</div>

<!-- Grid Cards -->
<div class="row row-cols-1 row-cols-md-3 g-4">
  <div class="col">
    <div class="card h-100">
      <div class="card-body">
        <h5 class="card-title">Card 1</h5>
        <p class="card-text">This is a longer card with supporting text.</p>
      </div>
    </div>
  </div>
  <div class="col">
    <div class="card h-100">
      <div class="card-body">
        <h5 class="card-title">Card 2</h5>
        <p class="card-text">This is a longer card with supporting text.</p>
      </div>
    </div>
  </div>
</div>
</div>

```

8. Grid Component

Grid System Basics:

html

```

<!-- Basic Grid -->
<div class="container">
  <div class="row">
    <div class="col">Column 1</div>
    <div class="col">Column 2</div>
    <div class="col">Column 3</div>
  </div>
</div>

```

```

<!-- Responsive Grid -->
<div class="container">
  <div class="row">
    <div class="col-sm-6 col-md-4 col-lg-3">
      <div class="p-3 border bg-light">Responsive Column</div>
    </div>
    <div class="col-sm-6 col-md-4 col-lg-3">
      <div class="p-3 border bg-light">Responsive Column</div>
    </div>
    <div class="col-sm-6 col-md-4 col-lg-3">
      <div class="p-3 border bg-light">Responsive Column</div>
    </div>
    <div class="col-sm-6 col-md-4 col-lg-3">
      <div class="p-3 border bg-light">Responsive Column</div>
    </div>
  </div>
</div>

<!-- Auto Layout -->
<div class="container">
  <div class="row">
    <div class="col">1 of 3</div>
    <div class="col-6">2 of 3 (wider)</div>
    <div class="col">3 of 3</div>
  </div>
</div>

<!-- Nesting Grids -->
<div class="container">
  <div class="row">
    <div class="col-8">
      Level 1: .col-8
      <div class="row">
        <div class="col-6">Level 2: .col-6</div>
        <div class="col-6">Level 2: .col-6</div>
      </div>
    </div>
  </div>
</div>

```

9. Grid Vertical Alignment

Vertical Alignment:

html

```
<div class="container">
  <!-- Align items start -->
  <div class="row align-items-start" style="height: 100px;">
    <div class="col">Top aligned column</div>
    <div class="col">Top aligned column</div>
  </div>

  <!-- Align items center -->
  <div class="row align-items-center" style="height: 100px;">
    <div class="col">Center aligned column</div>
    <div class="col">Center aligned column</div>
  </div>

  <!-- Align items end -->
  <div class="row align-items-end" style="height: 100px;">
    <div class="col">Bottom aligned column</div>
    <div class="col">Bottom aligned column</div>
  </div>

  <!-- Individual column alignment -->
  <div class="row" style="height: 100px;">
    <div class="col align-self-start">Top</div>
    <div class="col align-self-center">Center</div>
    <div class="col align-self-end">Bottom</div>
  </div>

  <!-- Vertical centering -->
  <div class="container min-vh-100 d-flex align-items-center justify-content-center">
    <div class="text-center">
      <h1>Vertically Centered</h1>
      <p>This content is vertically and horizontally centered</p>
    </div>
  </div>
</div>
```

10. Form Control Bootstrap

Basic Form Controls:

html

```
<form>

  <!-- Text Input -->
  <div class="mb-3">
    <label for="exampleInput" class="form-label">Text Input</label>
    <input type="text" class="form-control" id="exampleInput">
  </div>

  <!-- Email Input -->
  <div class="mb-3">
    <label for="exampleEmail" class="form-label">Email address</label>
    <input type="email" class="form-control" id="exampleEmail">
    <div class="form-text">We'll never share your email.</div>
  </div>

  <!-- Password -->
  <div class="mb-3">
    <label for="examplePassword" class="form-label">Password</label>
    <input type="password" class="form-control" id="examplePassword">
  </div>

  <!-- Textarea -->
  <div class="mb-3">
    <label for="exampleTextarea" class="form-label">Textarea</label>
    <textarea class="form-control" id="exampleTextarea" rows="3"></textarea>
  </div>

  <!-- Form Validation -->
  <div class="mb-3">
    <label for="validationInput" class="form-label">Required Field</label>
    <input type="text" class="form-control is-valid" id="validationInput" require
d>

    <div class="valid-feedback">Looks good!</div>
  </div>

  <div class="mb-3">
    <label for="invalidInput" class="form-label">Invalid Field</label>
    <input type="text" class="form-control is-invalid" id="invalidInput" required
>

    <div class="invalid-feedback">Please provide a valid value.</div>
  </div>
```

```

<!-- Form Sizes -->
<input class="form-control form-control-lg" type="text" placeholder="Large">
<input class="form-control" type="text" placeholder="Default">
<input class="form-control form-control-sm" type="text" placeholder="Small">

<!-- Disabled/Readonly -->
<input class="form-control" type="text" placeholder="Disabled" disabled>
<input class="form-control" type="text" value="Readonly" readonly>

<button type="submit" class="btn btn-primary">Submit</button>
</form>

```

11. Select in Form

Select Dropdowns:

html

```

<form>
  <!-- Basic Select -->
  <div class="mb-3">
    <label for="basicSelect" class="form-label">Basic Select</label>
    <select class="form-select" id="basicSelect">
      <option selected>Open this select menu</option>
      <option value="1">One</option>
      <option value="2">Two</option>
      <option value="3">Three</option>
    </select>
  </div>

  <!-- Sizing -->
  <div class="mb-3">
    <label for="largeSelect" class="form-label">Large Select</label>
    <select class="form-select form-select-lg" id="largeSelect">
      <option>Large select</option>
      <option>Option 1</option>
    </select>
  </div>

  <div class="mb-3">
    <label for="smallSelect" class="form-label">Small Select</label>

```

```

<select class="form-select form-select-sm" id="smallSelect">
  <option>Small select</option>
  <option>Option 1</option>
</select>
</div>

<!-- Multiple Select -->
<div class="mb-3">
  <label for="multipleSelect" class="form-label">Multiple Select</label>
  <select class="form-select" id="multipleSelect" multiple>
    <option selected>Option 1</option>
    <option>Option 2</option>
    <option>Option 3</option>
    <option>Option 4</option>
  </select>
</div>

<!-- Disabled Select -->
<div class="mb-3">
  <label for="disabledSelect" class="form-label">Disabled Select</label>
  <select class="form-select" id="disabledSelect" disabled>
    <option>Disabled select</option>
  </select>
</div>
</form>

```

12. Checkbox and Radio

Checkboxes and Radios:

html

```

<form>
  <!-- Checkboxes -->
  <div class="mb-3">
    <label class="form-label">Checkboxes</label>
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="check1">
      <label class="form-check-label" for="check1">Option 1</label>
    </div>
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="check2">

```

```

        <label class="form-check-label" for="check2">Option 2</label>
    </div>
</div>

<!-- Radio Buttons -->
<div class="mb-3">
    <label class="form-label">Radio Buttons</label>
    <div class="form-check">
        <input class="form-check-input" type="radio" name="radioGroup" id="radio
1">

        <label class="form-check-label" for="radio1">Radio 1</label>
    </div>
    <div class="form-check">
        <input class="form-check-input" type="radio" name="radioGroup" id="radio
2">

        <label class="form-check-label" for="radio2">Radio 2</label>
    </div>
</div>

<!-- Inline Checkboxes -->
<div class="mb-3">
    <label class="form-label">Inline Checkboxes</label>
    <div class="form-check form-check-inline">
        <input class="form-check-input" type="checkbox" id="inlineCheck1">
        <label class="form-check-label" for="inlineCheck1">1</label>
    </div>
    <div class="form-check form-check-inline">
        <input class="form-check-input" type="checkbox" id="inlineCheck2">
        <label class="form-check-label" for="inlineCheck2">2</label>
    </div>
</div>

<!-- Switches -->
<div class="mb-3">
    <div class="form-check form-switch">
        <input class="form-check-input" type="checkbox" id="flexSwitch">
        <label class="form-check-label" for="flexSwitch">Switch checkbox input</l
abel>
    </div>
</div>

<!-- Disabled States -->
<div class="form-check">
    <input class="form-check-input" type="checkbox" id="disabledCheck" disabled>

```

```

    <label class="form-check-label" for="disabledCheck">Disabled checkbox</label>
  </div>
</form>

```

13. Checkbox and Radio in Form

Integrated Form Example:

html

```

<form class="p-4 border rounded">
  <h4 class="mb-3">Registration Form</h4>

  <!-- Personal Information -->
  <div class="row mb-3">
    <div class="col-md-6">
      <label for="firstName" class="form-label">First Name</label>
      <input type="text" class="form-control" id="firstName" required>
    </div>
    <div class="col-md-6">
      <label for="lastName" class="form-label">Last Name</label>
      <input type="text" class="form-control" id="lastName" required>
    </div>
  </div>

  <!-- Contact Information -->
  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" class="form-control" id="email" required>
  </div>

  <!-- Gender Selection (Radio) -->
  <div class="mb-3">
    <label class="form-label">Gender</label>
    <div class="form-check">
      <input class="form-check-input" type="radio" name="gender" id="male">
      <label class="form-check-label" for="male">Male</label>
    </div>
    <div class="form-check">
      <input class="form-check-input" type="radio" name="gender" id="female">
      <label class="form-check-label" for="female">Female</label>
    </div>
  </div>

```



```

    <div class="form-check">
      <input class="form-check-input" type="radio" name="gender" id="other">
      <label class="form-check-label" for="other">Other</label>
    </div>
  </div>

  <!-- Interests (Checkboxes) -->
  <div class="mb-3">
    <label class="form-label">Interests</label>
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="sports">
      <label class="form-check-label" for="sports">Sports</label>
    </div>
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="music">
      <label class="form-check-label" for="music">Music</label>
    </div>
    <div class="form-check">
      <input class="form-check-input" type="checkbox" id="reading">
      <label class="form-check-label" for="reading">Reading</label>
    </div>
  </div>

  <!-- Country Select -->
  <div class="mb-3">
    <label for="country" class="form-label">Country</label>
    <select class="form-select" id="country" required>
      <option value="" selected disabled>Select country</option>
      <option value="us">United States</option>
      <option value="uk">United Kingdom</option>
      <option value="ca">Canada</option>
    </select>
  </div>

  <!-- Terms Agreement -->
  <div class="mb-3 form-check">
    <input class="form-check-input" type="checkbox" id="terms" required>
    <label class="form-check-label" for="terms">I agree to the terms and conditio
ns</label>
  </div>

  <!-- Submit Button -->

```

```
<button type="submit" class="btn btn-primary">Submit</button>
</form>
```

14. Form Layout

Form Layout Options:

html

```
<!-- Basic Form Layout -->
<form class="container mt-4">
  <h3 class="mb-4">Form Layout Examples</h3>

  <!-- Horizontal Form -->
  <div class="row mb-3">
    <label for="horizontalEmail" class="col-sm-2 col-form-label">Email</label>
    <div class="col-sm-10">
      <input type="email" class="form-control" id="horizontalEmail">
    </div>
  </div>

  <div class="row mb-3">
    <label for="horizontalPassword" class="col-sm-2 col-form-label">Password</label>
    <div class="col-sm-10">
      <input type="password" class="form-control" id="horizontalPassword">
    </div>
  </div>
</form>

<!-- Inline Form -->
<form class="row g-3 align-items-center mb-4">
  <div class="col-auto">
    <label class="visually-hidden" for="inlineUsername">Username</label>
    <input type="text" class="form-control" id="inlineUsername" placeholder="User
name">
  </div>
  <div class="col-auto">
    <label class="visually-hidden" for="inlinePassword">Password</label>
    <input type="password" class="form-control" id="inlinePassword" placeholder
="Password">
  </div>
```

```

<div class="col-auto">
  <button type="submit" class="btn btn-primary">Submit</button>
</div>
</form>

<!-- Grid-based Form Layout -->
<form class="container">
  <div class="row">
    <div class="col-md-6 mb-3">
      <label for="gridFirstName" class="form-label">First Name</label>
      <input type="text" class="form-control" id="gridFirstName">
    </div>
    <div class="col-md-6 mb-3">
      <label for="gridLastName" class="form-label">Last Name</label>
      <input type="text" class="form-control" id="gridLastName">
    </div>
  </div>

  <div class="row">
    <div class="col-md-8 mb-3">
      <label for="gridAddress" class="form-label">Address</label>
      <input type="text" class="form-control" id="gridAddress">
    </div>
    <div class="col-md-4 mb-3">
      <label for="gridZip" class="form-label">Zip</label>
      <input type="text" class="form-control" id="gridZip">
    </div>
  </div>
</form>

<!-- Form with Validation States -->
<form class="was-validated">
  <div class="mb-3">
    <label for="validatedInput" class="form-label">Validated Input</label>
    <input type="text" class="form-control" id="validatedInput" required>
    <div class="valid-feedback">Looks good!</div>
    <div class="invalid-feedback">Please provide a valid input.</div>
  </div>
</form>

```

Complete Example Page

Here's a complete HTML page combining all concepts:

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html

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Bootstrap Complete Guide</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
  <style>
    .component-section {
      border: 1px solid #dee2e6;
      border-radius: 0.375rem;
      padding: 1.5rem;
      margin-bottom: 2rem;
    }
    .component-title {
      color: #0d6efd;
      border-bottom: 2px solid #0d6efd;
      padding-bottom: 0.5rem;
      margin-bottom: 1rem;
    }
    .example-box {
      background-color: #f8f9fa;
      padding: 1rem;
      border-radius: 0.375rem;
      margin: 1rem 0;
    }
  </style>
</head>
<body>
  <div class="container py-4">
    <h1 class="text-center mb-4">Bootstrap Complete Guide</h1>

    <!-- Navbar Example -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-primary mb-4">
      <div class="container-fluid">
        <a class="navbar-brand" href="#">Bootstrap Guide</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse"
          data-bs-target="#navbarContent">
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        <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarContent">
        <ul class="navbar-nav me-auto">
            <li class="nav-item"><a class="nav-link active" href="#">Home
</a></li>

            <li class="nav-item"><a class="nav-link" href="#">Components
</a></li>

            <li class="nav-item"><a class="nav-link" href="#">Forms</a></
li>

        </ul>
    </div>
</div>
</nav>

<!-- Grid System Example -->
<div class="component-section">
    <h3 class="component-title">Grid System</h3>
    <div class="row">
        <div class="col-md-4">
            <div class="example-box">
                <h5>Column 1</h5>
                <p>Responsive column that becomes full width on mobile.</p>
            </div>
        </div>
        <div class="col-md-4">
            <div class="example-box">
                <h5>Column 2</h5>
                <p>Uses Bootstrap's 12-column grid system.</p>
            </div>
        </div>
        <div class="col-md-4">
            <div class="example-box">
                <h5>Column 3</h5>
                <p>Automatically adjusts for different screen sizes.</p>
            </div>
        </div>
    </div>
</div>

<!-- Buttons and Badges -->
<div class="component-section">
    <h3 class="component-title">Buttons & Badges</h3>
    <div class="example-box">

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<button class="btn btn-primary me-2">Primary</button>
<button class="btn btn-success me-2">
    Success <span class="badge bg-light text-dark">4</span>
</button>
<button class="btn btn-danger position-relative me-2">
    Danger
    <span class="position-absolute top-0 start-100 translate-middle b
badge rounded-pill bg-dark">
        9+
    </span>
</button>
<span class="badge bg-warning text-dark">Warning Badge</span>
</div>
</div>

<!-- Complete Form Example -->
<div class="component-section">
    <h3 class="component-title">Complete Form Example</h3>
    <form class="row g-3">
        <div class="col-md-6">
            <label class="form-label">Full Name</label>
            <input type="text" class="form-control" placeholder="Enter your n
ame">
        </div>
        <div class="col-md-6">
            <label class="form-label">Email</label>
            <input type="email" class="form-control" placeholder="name@exampl
e.com">
        </div>
        <div class="col-12">
            <label class="form-label">Subscription Type</label>
            <div class="form-check">
                <input class="form-check-input" type="radio" name="subscripti
on" id="basic">
                <label class="form-check-label" for="basic">Basic</label>
            </div>
            <div class="form-check">
                <input class="form-check-input" type="radio" name="subscripti
on" id="premium">
                <label class="form-check-label" for="premium">Premium</label>
            </div>
        </div>
        <div class="col-12">
            <label class="form-label">Features</label>

```

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1">
    <input class="form-check-input" type="checkbox" id="feature
    <label class="form-check-label" for="feature1">Feature 1</lab
el>

</div>
<div class="form-check">
    <input class="form-check-input" type="checkbox" id="feature
2">
    <label class="form-check-label" for="feature2">Feature 2</lab
el>

</div>
</div>
<div class="col-md-6">
    <label class="form-label">Country</label>
    <select class="form-select">
        <option selected>Select country</option>
        <option>United States</option>
        <option>Canada</option>
        <option>United Kingdom</option>
    </select>
</div>
<div class="col-md-6">
    <label class="form-label">Plan</label>
    <select class="form-select">
        <option selected>Choose plan</option>
        <option>Basic - $9.99/month</option>
        <option>Pro - $19.99/month</option>
        <option>Enterprise - $49.99/month</option>
    </select>
</div>
<div class="col-12">
    <div class="form-check">
        <input class="form-check-input" type="checkbox" id="terms">
        <label class="form-check-label" for="terms">Agree to terms</l
abel>

    </div>
</div>
<div class="col-12">
    <button type="submit" class="btn btn-primary">Submit Form</button>

    <button type="reset" class="btn btn-secondary">Reset</button>
</div>
</form>

```

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</div>

<!-- Alert Example -->
<div class="component-section">
  <h3 class="component-title">Alert Component</h3>
  <div class="alert alert-success alert-dismissible fade show">
    <strong>Success!</strong> Form submitted successfully.
    <button type="button" class="btn-close" data-bs-dismiss="alert"></but
ton>
  </div>
</div>

<!-- Card Example -->
<div class="component-section">
  <h3 class="component-title">Card Component</h3>
  <div class="row">
    <div class="col-md-6">
      <div class="card">
        <div class="card-header bg-primary text-white">
          <h5 class="mb-0">Featured Card</h5>
        </div>
        <div class="card-body">
          <p class="card-text">This is an example of a Bootstrap ca
rd component with header, body, and footer.</p>
          <a href="#" class="btn btn-primary">Learn More</a>
        </div>
        <div class="card-footer text-muted">
          Last updated 2 days ago
        </div>
      </div>
    </div>
  </div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Key Takeaways:

1. **Bootstrap provides ready-to-use components** that save development time
2. **Responsive design** is built-in with the grid system
3. **Consistent styling** across all elements and browsers
4. **Mobile-first approach** ensures compatibility on all devices
5. **Extensive documentation** and community support
6. **Customizable** through SASS variables
7. **Accessibility features** built into components

This guide covers the essential Bootstrap components you'll need for most web projects. Remember to always check the [official Bootstrap documentation](#) for the latest features and detailed examples.